Dynamics II

$$\bar{F}_{tot} = m\bar{a} = m\frac{d\bar{v}}{dt} = m\frac{d^2\bar{r}}{dt^2}$$

$$\bar{F}_{tot} = m\bar{g} + \bar{F}_A + \underline{\mu N \hat{e}_v} + \frac{k q_1 q_2}{r^2}\hat{e}_r + \frac{G m_1 m_2}{r^2}\hat{e}_r - k\Delta r\,\hat{e}_r + \underline{\bar{F}_{viscr}} + \bar{T} + \dots$$

$$\updownarrow ?$$

$$U = mgr - \frac{k q_1 q_2}{r} - \frac{G m_1 m_2}{r} + \frac{k\Delta r^2}{2} \quad - \text{potential energy}$$

$$\bar{F} = -\frac{dU}{dr}\hat{e}_r \quad - \text{conservative forces}$$



Work – Energy

$\bar{v} \cdot \bar{v}$ $\quad \bar{F} = m\bar{a}$

$$\bar{F}_{tot} \cdot d\bar{r} = dK, \quad K = \frac{mv^2}{2}$$

$$\int_{x_1}^{x_2} f'(x)\,dx = f(x_2) - f(x_1)$$

$$\int_{x_1}^{x_1} df(x) = \Delta f$$

df — total derivative

} Fundamental theorem of Calculus

$d \to \lim_{\Delta \to 0} \Delta$

1) $\bar{N} \cdot d\bar{r} = N \cdot dr \cos 90 = 0$ ✓

2) $-\mu N \bar{e}_v \cdot d\bar{r} = -\mu N\, dr$

$\bar{F}_{dr} = -\lambda v^2 \hat{e}_v$ ✗

3) $\bar{F}_{cons} \cdot d\bar{r} = -\frac{du}{dr} \hat{e}_r \cdot d\bar{r} = -\frac{du}{dr} dr = -du$ ✓

$-du = dK$

$0 = dK + du = d(K+U)$

$K + U = const$ — Energy conservation $\Rightarrow K_1 + U_1 = K_2 + U_2$

Impulse — Momentum

$$\bar{F} = m\bar{a}$$

$$\bar{F} = m\frac{d\bar{v}}{dt}$$

$$\bar{F}\cdot dt = m\,d\bar{v} = d\,m\bar{v}$$

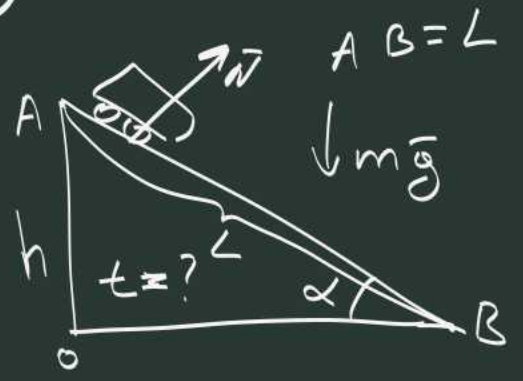
$\bar{p} = m\bar{v}$ – linear momentum

$$\bar{J} = \int \bar{F}\cdot dt = \int_{\bar{p}_1}^{\bar{p}_2} d\bar{p} = \bar{p}_2 - \bar{p}_1 = \Delta\bar{p}$$

Impulse

$$W = \Delta K$$
$$\bar{J} = \Delta\bar{p}$$

Example 1



$$\bar{v} = \bar{v}_0 + \bar{a}\cdot t$$

$$\bar{v} = \bar{a}\,t =$$

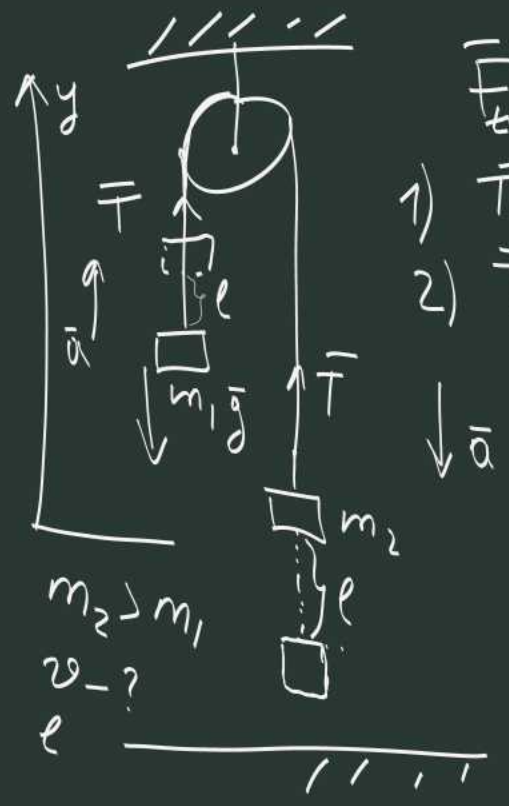
$$v = g\sin\alpha \cdot t$$

$$0 + mgh = \frac{mv^2}{2} + 0$$

$$2gh = v^2 = g^2\sin^2\alpha \cdot t^2$$

$$2L\sin\alpha = g\sin^2\alpha \cdot t^2 \Rightarrow t = \sqrt{\frac{2L}{g\sin\alpha}}$$

Example 2



$$\bar{F}_{tot} = m\bar{a}$$

1) $\bar{T} + m_1\bar{g} = m_1\bar{a}$

2) $\bar{T} + m_2\bar{g} = m_2\bar{a}$

y:

1) $T - m_1 g = m_1 a$

2) $T - m_2 g = -m_2 a$

$$m_2 g - m_1 g = (m_1 + m_2)a$$

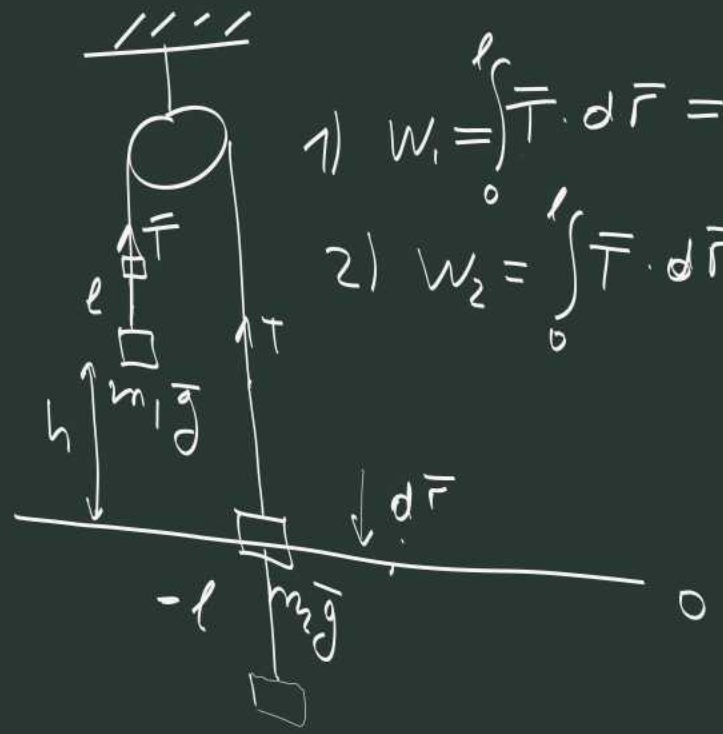
$$a = \frac{(m_2 - m_1)g}{m_1 + m_2}$$

$$\ell = \frac{at^2}{2} \Rightarrow t = \sqrt{\frac{2\ell}{a}}$$

$$v = at = \sqrt{2\ell a}$$

$$\boxed{v = \sqrt{\frac{2\ell(m_2 - m_1)g}{m_1 + m_2}}}$$

$\Rightarrow$



1) $W_1 = \int_0^l \bar{T} \cdot d\bar{r} = T \cdot l$

2) $W_2 = \int_0^l \bar{T} \cdot d\bar{r} = -\int_0^l T\, dr = -T \cdot l$

$$m_1 g h = m_1 g (h + l) + \frac{m_1 v^2}{2} + \frac{m_2 v^2}{2} - m_2 g l$$

$$0 = \underline{m_1 g l} + \frac{m_1 + m_2}{2} v^2 - \underline{m_2 g l}$$

$$0 = (m_1 - m_2) g l + \frac{m_1 + m_2}{2} v^2$$

$$(m_2 - m_1) g l = \frac{m_1 + m_2}{2} v^2$$

$$v = \sqrt{\frac{2 (m_2 - m_1) g l}{m_1 + m_2}}$$